Capacity Report

Monthly close 2026-10-23 · Sample · vc-lab01

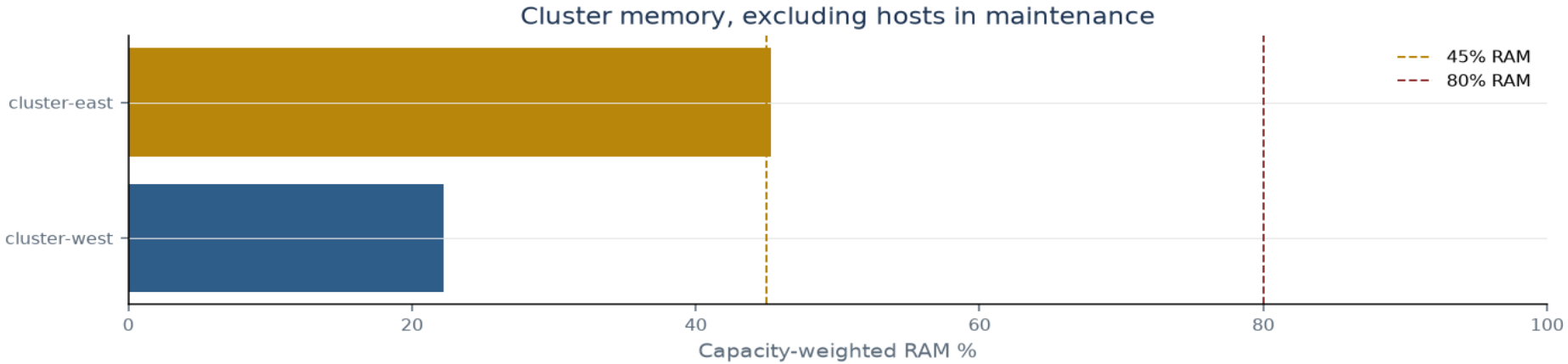
Powered-on VMs	Hosts	Memory used	CPU weighted	Datastore free	Status
53	**6 (1 maint.)**	**31.0%**	**15.2%**	**36.1%**	**AMBER**

Executive summary

- Overall status AMBER as of 2026-10-23 across 1 vCenter(s): vc-lab01.
- 1 datastore(s) below 20% free
- 53 powered-on VMs (54 excluding templates) on 6 hosts (1 in maintenance).
- Memory 1431 / 4608 GiB (31.0%). CPU 15.2% weighted by cores (simple average 15.2%).
- Shared datastore 19.2 / 30.0 TiB used (36.1% free).
- Trend uses 8 weekly snapshots; linear forecast is 12 weeks ahead.

1. Host CPU and memory

Watch lines: 45% RAM and 45% CPU. Critical is 80%. Balloon/swap is the real RAM distress signal.

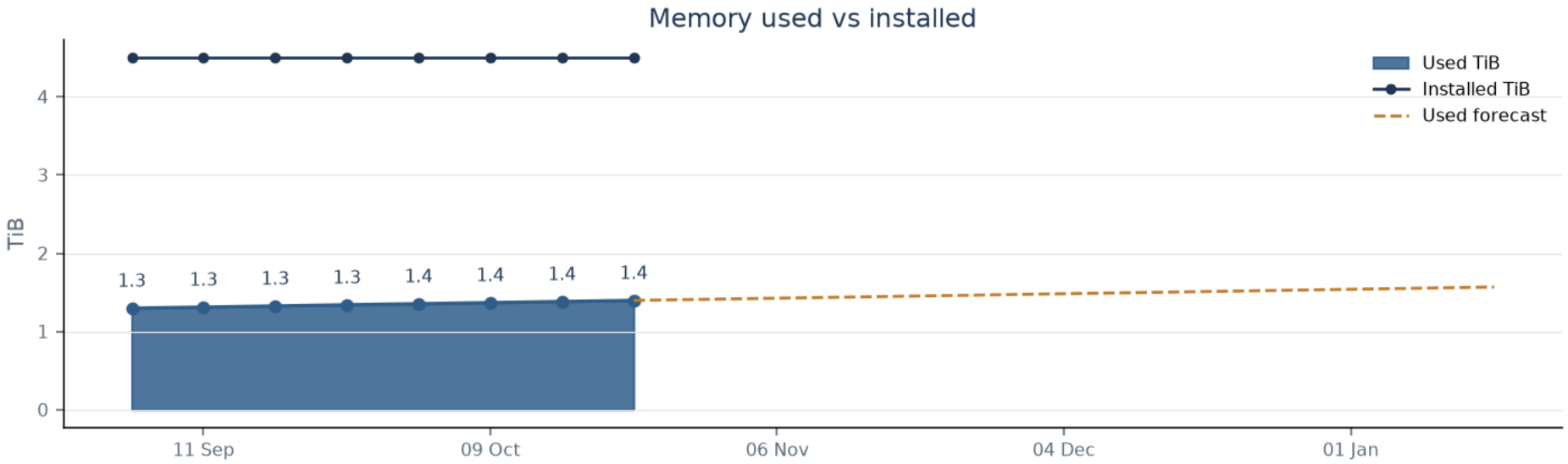


Host	Cluster	Maint	CPU %	Mem %	Mem GiB	Cores	vCPU/core	VMs
esx-e2.lab.example	cluster-east		26%	52%	768	32	4.25	11
esx-e3.lab.example	cluster-east		16%	46%	768	32	2.25	11
esx-e1.lab.example	cluster-east		20%	38%	768	32	2.00	11

Host	Cluster	Maint	CPU %	Mem %	Mem GiB	Cores	vCPU/core	VMs
esx-w1.lab.example	cluster-west		12%	24%	768	32	2.00	10
esx-w2.lab.example	cluster-west		14%	20%	768	32	2.00	10
esx-w3.lab.example	cluster-west	Yes	3%	6%	768	32	0.50	0

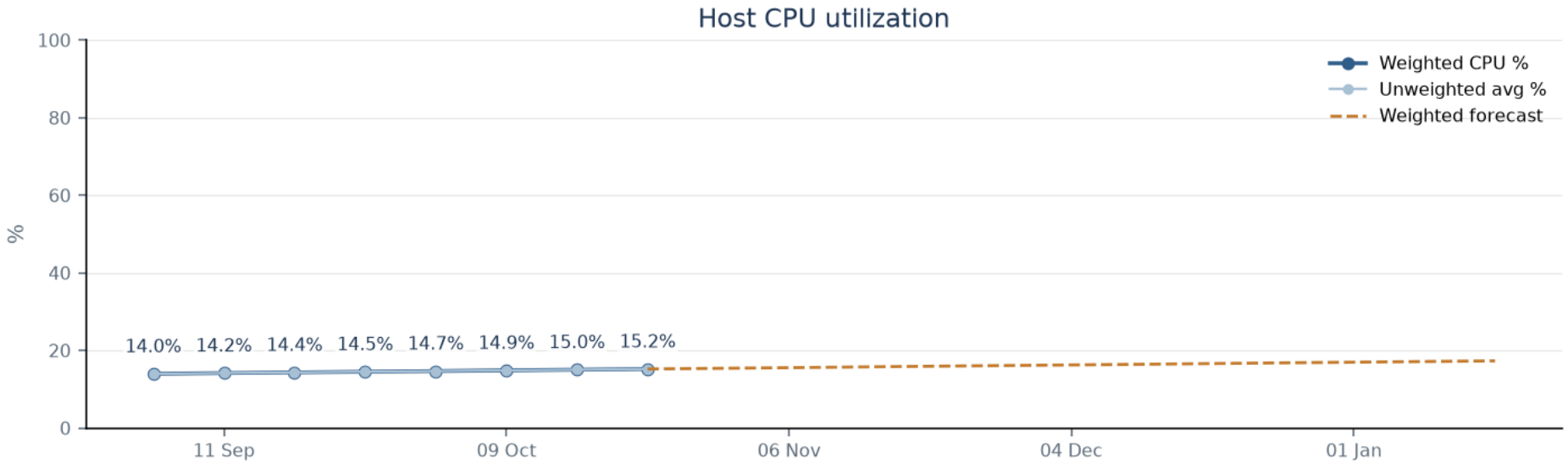
- 2 host(s) at or above 45% RAM: esx-e2.lab.example, esx-e3.lab.example.
- No host above the 45% CPU watch line.
- 1 host(s) in maintenance — exclude them from spare-capacity claims.
- No balloon or swap in vMemory — hosts are not in RAM distress even if some sit above 45%.
- 1 host(s) at ≥ 4.0 vCPU/core. That is consolidation, not necessarily CPU saturation.

2. Memory trend and forecast



- Installed memory is 4608 GiB with 31.0% utilized (capacity-weighted from host counters).
- Used RAM changed +15 GiB versus 2026-10-16 (1416 → 1431).
- Linear forecast of used RAM in 12 weeks: 1606 GiB. Treat as a trend, not a booking.

3. CPU trend and forecast



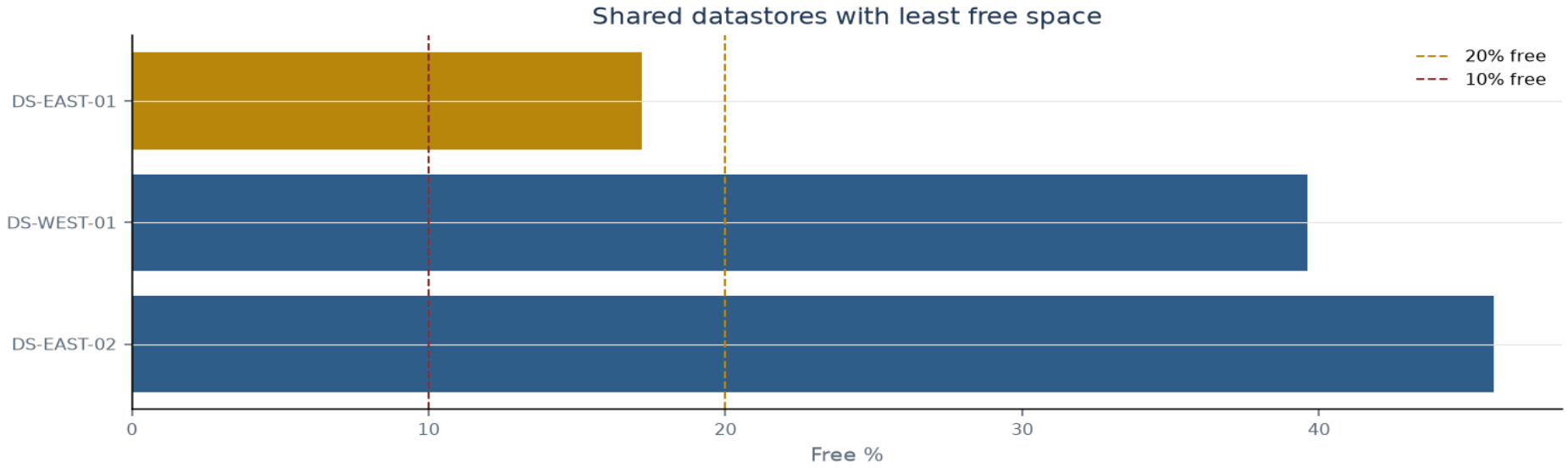
- Capacity-weighted CPU is 15.2% (better than averaging host % columns, which is 15.2% here).
- Linear forecast of weighted CPU in 12 weeks: 17.3%.

Weekly snapshots

Week	VMs	Δ VMs	RAM GiB	Δ RAM	CPU %	DS free %	Snapshots
2026-09-04	46		1329		14.0%	40.1%	3
2026-09-11	47	+1	1343	+15	14.2%	39.5%	4
2026-09-18	48	+1	1358	+15	14.4%	38.9%	5
2026-09-25	49	+1	1372	+15	14.5%	38.4%	6
2026-10-02	50	+1	1387	+15	14.7%	37.8%	7
2026-10-09	51	+1	1402	+15	14.9%	37.3%	8
2026-10-16	52	+1	1416	+15	15.0%	36.7%	9
2026-10-23	53	+1	1431	+15	15.2%	36.1%	9

4. Datastore space

Host-local, depot and ISO datastores are omitted so 99% free scratch LUNs do not look like stranded capacity.



Datastore	Cluster	VMs	Cap TiB	Prov TiB	Used TiB	Free TiB	Free %
DS-EAST-01	cluster-east	18	8.00	6.96	6.62	1.38	17%
DS-WEST-01	cluster-west	14	10.00	6.34	6.04	3.96	40%
DS-EAST-02	cluster-east	16	12.00	13.80	6.49	5.51	46%

- Tightest shared datastore is DS-EAST-01 at 17% free (1.4 TiB of 8.0 TiB).
- 1 datastore(s) below 20% free — plan growth or cleanup.
- 1 datastore(s) are thin-overprovisioned (provisioned > capacity). Watch in-use, not provisioned.

5. Guest filesystem hotspots

Critical ≤ 10% free. Warning ≤ 20% free. /boot and volumes under 1 GiB are hidden.

Critical

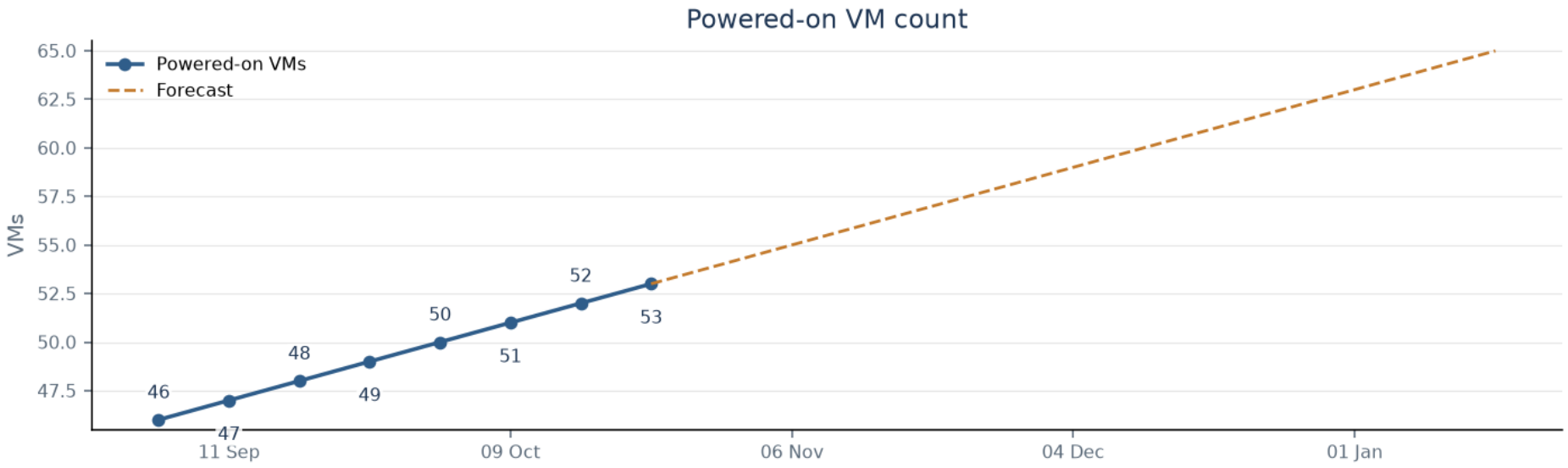
VM	Disk	Cap GiB	Used GiB	Free GiB	Free %
app-db-01	/var	20	19	1	7%
app-web-04	/var	12	11	1	8%

Warning (includes critical)

VM	Disk	Cap GiB	Used GiB	Free GiB	Free %
app-db-01	/var	20	19	1	7%
app-web-04	/var	12	11	1	8%
app-api-02	/var	16	14	2	14%

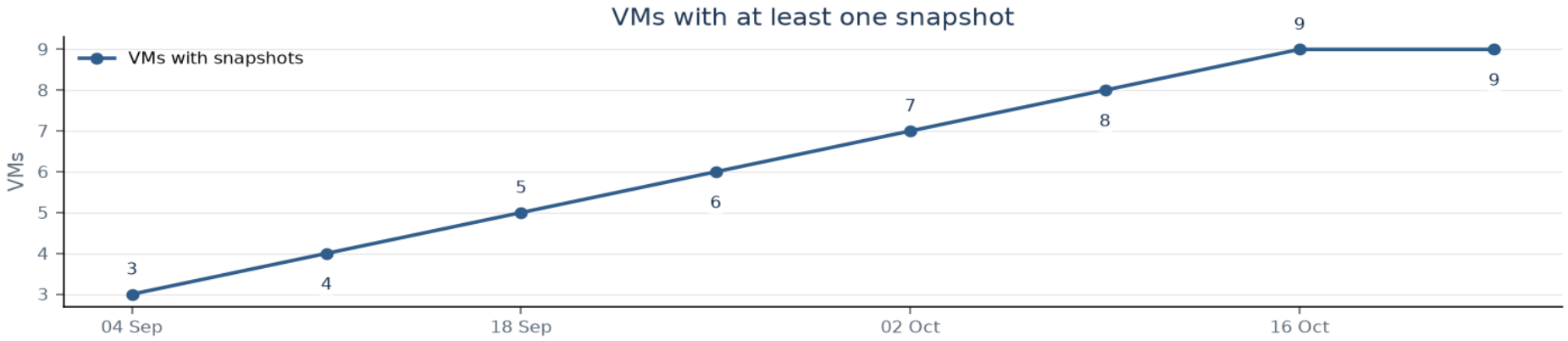
- Tiny volumes (/boot, < 1 GiB) are excluded so the list is not dominated by small system mounts.
- 2 guest filesystem(s) ≤ 10% free. Worst: app-db-01 /var (7%), app-web-04 /var (8%).
- 3 guest filesystem(s) ≤ 20% free including the critical set.

6. Powered-on VM count



- 53 powered-on VMs this snapshot (2 templates excluded from the count).
- Powered-on count +1 versus 2026-10-16.

7. Hygiene



- 9 VM(s) still have snapshots (9 snapshot files). Age them out unless they are active change windows.
- 2 VM(s) with Tools not running or not installed.